

Full production three-component gradient flow: canonical theorem route

TECT verification-first repository · claim A2-FULL-PRODUCTION-WELLPOSED

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1. Purpose and scope

This note starts P2 without changing the older scalar claim **A2-PDE-WELLPOSED**. Its object is the real- L^2 gradient flow of the canonical three-component functional fixed by **A1-PRODUCTION-FUNCTIONAL-REALISATION**. The historical external backend is not used: P1 records it as a non-variational proxy. The actual production value $\eta_{\text{shell}} = 0$ is imposed. A nonzero shell-bias continuum limit is outside this note because its discrete normalisation has not been identified with a continuum functional.

The target theorem is deliberately an H^2 theorem. For $\Psi_0 \in H^2(\mathbb{T}^3; \mathbb{C}^3)$, the candidate conclusion is a unique global solution, continuous dependence in H^2 , monotone energy, and smoothness for positive time. Sections 2–4 close the algebraic and coercive baseline. Sections 5–7 assemble the analytic route but remain subject to the three named independent-audit gates on the claim card. Consequently this note is T4, not a T6 theorem enactment.

2. Canonical functional and real Hilbert structure

Let

$$\mathcal{H} = L^2(\mathbb{T}^3; \mathbb{C}^3), \quad \langle U, V \rangle_{\mathbb{R}} = \text{Re} \int_{\mathbb{T}^3} U^\dagger V \, dx.$$

Write $\rho = \Psi^\dagger \Psi$, and let T_A be the three Hermitian generators fixed by P1. Define

$$m_A = \Psi^\dagger T_A \Psi, \quad J_A = \nabla m_A, \quad q_A = \frac{m_A}{\rho + \epsilon_\rho}, \quad K_A = J_A - q_A \nabla \rho,$$

where $\epsilon_\rho = 10^{-12}$. The canonical functional is

$$\mathcal{F}[\Psi] = \frac{1}{2} \langle \Psi, L\Psi \rangle_{\mathbb{R}} + \int_{\mathbb{T}^3} \left(\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \right) dx \quad (1)$$

$$+ \sum_A \int_{\mathbb{T}^3} \left(\frac{a}{2} |J_A|^2 + b \text{Re}(J_A^\dagger K_A) + \frac{c}{2} |K_A|^2 \right) dx. \quad (2)$$

Here

$$L = (r + Z(-\Delta) + Y(-\Delta)^2)I_3 + M_{\text{fam}} + k_{\text{lock}}(I_3 - P_0),$$

$M_{\text{fam}} = \text{diag}(0, 0.03, 0.07)$, $P_0 = z_0 z_0^\dagger / (z_0^\dagger z_0)$, and $k_{\text{lock}} = 0.15$. The Class-II denominator is $M_X^2 + \epsilon_M$ with $\epsilon_M = 10^{-12}$, and

$$a = \frac{c_{JJ}\alpha_X^2}{M_X^2 + \epsilon_M}, \quad b = \frac{c_{JK}\alpha_X\beta_X}{M_X^2 + \epsilon_M}, \quad c = \frac{c_{KK}\beta_X^2}{M_X^2 + \epsilon_M}.$$

The PDE is the real gradient flow

$$\partial_t \Psi = -\nabla_{\mathbb{R}} \mathcal{F}[\Psi].$$

3. Full linear self-adjointness and lower bound

On a Fourier mode k , the scalar part of L has the real symbol

$$p(|k|^2) = r + Z|k|^2 + Y|k|^4, \quad r = 0.4740336473, \quad Z = -0.9252754126, \quad Y = 1.$$

Since $Y > 0$, the multiplier is self-adjoint on H^4 . Completing the square gives the continuum shell location and mass

$$q_{\text{sh}} = \sqrt{-Z/(2Y)} = 0.6801747615870498, \quad m_{\text{sh}}^2 = r - \frac{Z^2}{4Y} = 0.260000000009475 > 0.$$

Every discrete torus mode therefore satisfies $p(|k|^2) \geq m_{\text{sh}}^2$. More strongly,

$$c_{H^2} := \min_{x \geq 0} \frac{r + Zx + Yx^2}{1 + x^2} = 0.2048572626782363 > 0,$$

so the scalar quadratic form controls the H^2 norm. The family matrix is Hermitian positive semidefinite. The normalised P_0 is a Hermitian idempotent, hence $I - P_0$ is positive semidefinite; multiplication by the nonnegative lock coefficient preserves this sign. Bounded Hermitian internal terms preserve self-adjointness on H^4 and cannot reduce either lower bound. This closes proof stage 1 in the declared production scope.

4. Class-II sign and energy coercivity

For each generator and spatial direction the Class-II density is the quadratic form with coefficient matrix

$$Q_{II} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \begin{pmatrix} 0.004499999999998875 & 0.0018749999999995311 \\ 0.0018749999999995311 & 0.002343749999999414 \end{pmatrix}.$$

Its determinant and eigenvalues are

$$\det Q_{II} = 7.031249999996483 \times 10^{-6}, \quad \sigma(Q_{II}) = \{0.001259011500926061, 0.005584738499072228\}.$$

Thus the Class-II energy is nonnegative and coercive in the pair (J_A, K_A) . Because $\rho + \epsilon_\rho$ is strictly positive, no singular coefficient is introduced. This closes proof stage 2 for the canonical subset.

The production potential has $\lambda = -0.43$ and $\gamma = 1.62 > 0$. Pointwise,

$$\frac{|\lambda|}{4} \rho^2 \leq \frac{\gamma}{12} \rho^3 + C_Y, \quad q_{\text{quad}} C_Y = \frac{|\lambda|^3}{3\gamma^2} = 0.010098435197886498.$$

Combining this inequality with the linear and Class-II bounds yields

$$\mathcal{F}[\Psi] \geq \frac{c_{H^2}}{2} \|\Psi\|_{H^2}^2 + \frac{\gamma}{12} \int_{\mathbb{T}^3} \rho^3 dx - C_Y |\mathbb{T}^3|. \quad (4.1)$$

The deterministic audit reproduces all values above with 20 assertions and also verifies the P1 source hash, family/lock signs, projector identities, regularisers, and the zero shell-bias restriction.

5. Local existence and uniqueness: proof candidate

Identify $\mathbb{C}^3$ with $\mathbb{R}^6$ and write the field as u . The quantities m_A and ρ are quadratic in u . Hence

$$J_A = Dm_A(u) \nabla u, \quad K_A = (Dm_A(u) - q_A(u) D\rho(u)) \nabla u.$$

Since $\rho + \epsilon_\rho > 0$, every coefficient is a smooth function of u . The Class-II density therefore has the real-coordinate form

$$G_{II}(u, \nabla u) = \frac{1}{2} B_{\alpha\beta}^{j\ell}(u) \partial_j u_\alpha \partial_\ell u_\beta,$$

with smooth B . Its Euler–Lagrange map is a sum of terms of the two forms

$$B(u) \nabla^2 u, \text{quad} DB(u) [\nabla u, \nabla u]. \quad (5.1)$$

On an H^2 ball in three dimensions, u is bounded in L^∞ and ∇u is bounded in L^6 . Thus the first term of (5.1) lies in L^2 ; the second lies in L^3 , hence also in L^2 on the fixed finite torus. Applying the same product estimates to a difference gives the candidate bound

$$\|N_{II}(u) - N_{II}(v)\|_{L^2} \leq C_R \|u - v\|_{H^2}, \quad \|u\|_{H^2}, \|v\|_{H^2} \leq R. \quad (5.2)$$

The quartic and sextic Euler–Lagrange terms satisfy the same local-Lipschitz bound because H^2 is an algebra. Consequently the full lower-order map $N : H^2 \rightarrow L^2$ is locally Lipschitz, provided the coordinate reduction leading to (5.1) is confirmed term by term.

The positive self-adjoint fourth-order operator L generates an analytic semigroup and $D(L^{1/2}) = H^2$ with equivalent norms. Since

$$\|L^{1/2} e^{-tL}\|_{L^2 \rightarrow L^2} \leq C t^{-1/2},$$

the singularity is integrable. The standard contraction argument for

$$u(t) = e^{-tL} u_0 - \int_0^t e^{-(t-s)L} N(u(s)) ds$$

then gives a unique local mild solution in $C([0, T]; H^2)$ and the continuation alternative that a finite maximal time forces the H^2 norm to diverge. Equation (5.2) and its explicit coordinate expansion are the subject of **A2-FULL-NONLINEAR-MAPPING-AUDIT**; no theorem-tier claim is made before that audit closes.

6. Energy decrease and global continuation: proof candidate

Let P_n be the Fourier projection onto finitely many modes. The projected equation is a finite-dimensional real gradient flow for $\mathcal{F}(P_n u)$, so

$$\frac{d}{dt} \mathcal{F}(u_n(t)) = -\|\nabla_{\mathbb{R}} \mathcal{F}(u_n(t))\|_{L^2}^2 \leq 0. \quad (6.1)$$

The coercive estimate (4.1) bounds u_n uniformly in H^2 on every interval on which the initial energy is fixed. The functional is continuous on H^2 : the linear part is its quadratic norm, the local potential is continuous by $H^2 \hookrightarrow L^\infty$, and the Class-II density is quadratic in ∇u with smooth coefficients of u .

The candidate limiting argument uses the uniform H^2 bound, the projected equation, compact Sobolev embedding on the torus, and lower semicontinuity to pass (6.1) to the local solution. It yields

$$\mathcal{F}(u(t)) + \int_0^t \|\nabla_{\mathbb{R}} \mathcal{F}(u(s))\|_{L^2}^2 ds = \mathcal{F}(u_0). \quad (6.2)$$

Then (4.1) and (6.2) give a uniform H^2 bound. The local continuation alternative cannot fire, so the solution extends globally. The exact Class-II chain rule and the compactness passage in (6.2) remain the named **A2-FULL-ENERGY-CONTINUATION-AUDIT**.

7. Continuous dependence and smoothing: proof candidate

Local continuous dependence follows from (5.2) and the same weakly singular semigroup estimate used in the contraction. On a finite time interval the energy bound keeps solutions with nearby initial data in a common H^2 ball; iterating the local estimate gives continuous dependence for the global flow.

For every $t_0 > 0$, analytic-semigroup regularity first upgrades the solution to $D(L) = H^4$ on $[t_0, T]$. Smoothness of the regularised coefficient map and the fact that it is only second order then give

$$N : H^s \longrightarrow H^{s-2}$$

on bounded sets at higher Sobolev orders. Repeating the fourth-order semigroup gain bootstraps by two spatial derivatives and gives $u \in C^\infty((0, \infty) \times \mathbb{T}^3)$. The time-regularity input for the first H^4 gain and the induction are retained as **A2-FULL-SMOOTHING-AUDIT**.

Candidate theorem. For the canonical production functional, with $\eta_{\text{shell}} = 0$ and the pinned positive regularisers, every $u_0 \in H^2(\mathbb{T}^3; \mathbb{R}^6)$ generates a unique global H^2 solution of the real- L^2 gradient flow. The solution depends continuously on its initial datum, obeys (6.2), and is smooth for $t > 0$. This paragraph is the target assembled by the note, not a T6 enactment while the three audit gates remain open.

8. Devil's-advocate

- “*P1 T5 already proves the PDE theorem.*” UPHELD as an invalid reading. P1 closes a discrete variational matrix, not continuum evolution.
- “*The Class-II cross term may destroy the lower bound.*” DISMISSED at the production point: Q_{II} is positive definite with minimum eigenvalue 0.001259011500926061.
- “*The nonzero shell-bias test can be imported into the PDE.*” VALID with mitigation. Only the actual production value $\eta_{\text{shell}} = 0$ is included; nonzero normalisation and convergence are separate work.
- “*Writing a smooth coefficient $B(u)$ hides derivative loss.*” VALID as an audit demand. Formula (5.1) shows the proposed order count, but the complete six-real-component expansion must independently confirm it.
- “*Energy decrease is automatic for any formal gradient.*” UPHELD as an invalid shortcut. The Galerkin chain rule and limiting passage are explicitly retained as an open gate.
- “*Smoothing follows without a first positive-time domain gain.*” UPHELD as an invalid shortcut. The H^4 entry and induction must be audited.

Quantitative sanity check: the independent executable audit reports 20/20 PASS, $m_{\text{sh}}^2 = 0.260000000009475$, $c_{H^2} = 0.2048572626782363$, $\lambda_{\min}(Q_{II}) = 0.001259011500926061$, and $C_Y = 0.010098435197886498$.

9. Result footer

Result ID:	A2-FULL-PRODUCTION-WELLPOSED
Precise statement:	The production linear and Class-II terms are coercive; a global H2 proof candidate is assembled.
Scope:	Fixed T3, C3 field, canonical P1 functional, positive regularisers, eta_shell=0, H2 initial data.
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION.

Evidence grade: ANALYTIC + EXECUTED; T4 proof candidate.
Reproduction command: python
codes/foundations/
a2_full_production_wellposedness_checks.py
Expected output: 20/20 PASS;
A2-FULL-COERCIVITY-BASELINE-PASS; exit 0.
Falsification gate: Failed H2->L2 map, energy passage, continuation,
smoothing, or a counterexample in the stated scope.
Tier before / after: T0 / T4 FULL-PROOF-CANDIDATE.
No-overclaim statement: Not T6; not historical proxy; not eta_shell!=0;
not below H2, infinite volume, BCC, or stability.
Next required action: Audit the nonlinear map, energy continuation, and
smoothing gates in that order.